

Chapter XIII

Reliving History with “Reliving the Revolution”: Designing Augmented Reality Games to Teach the Critical Thinking of History

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Abstract

Students need to learn the critical thinking of history, yet they rarely have opportunities to authentically simulate historic inquiry. Research has suggested the pedagogical potential for using augmented reality (AR) games—location-based games that use wireless handheld devices such as PDAs to provide virtual game information in a physical environment. The novel AR game, Reliving the Revolution (RtR), was created as a model for studying how AR games can engage students in interpretive, collaborative, and problem-solving activities. In this chapter, the game is introduced, and main results of the initial iterative tests are discussed, including what went wrong and how the game was redesigned to better support deeper engagement and historical thinking and learning.

Introduction

There may be at least two versions to every story, but how do you determine the truth when both sides have valid, but differing, perspectives? Active participants in a democracy must be able to question sources, seek out and manage differing viewpoints, and develop their own interpretations of the information they receive. Social problems do not have one clear solution; rather they require the complex consideration of multiple possibilities, prior knowledge sets, and rubrics (Brush & Saye, 2005). Likewise, historians weigh evidence and decide to emphasize the particular perspectives that they feel are the best representations of the past. K-12 social studies students typically receive a litany of facts, events, names, along with one master narrative; they are rarely encouraged to empathize with alternate views or question the so-called authoritative versions of history. Teaching as though there is only one right way to view history is problematic because students are not practicing the skills necessary for historic inquiry (Hoge, 2003), and also because they are not learning how to unravel the complexity of social problems, nor evaluate the world as an engaged citizen. In this chapter, I present a new augmented reality game, *Reliving the Revolution* (RtR), as a model for teaching historic inquiry and critical thinking, and for considering how to design engaging educational games. RtR is not envisioned as a standalone educational solution, but as an activity supported by a teacher or mentor, and integrated into a broader history curriculum that incorporates experiential learning, teamwork, and critical thinking skills.

Overview of RtR

What better way to prepare students for skills essential to democratic engagement than by immersing them in a time when these democratic values were being questioned? RtR takes place in Lexington, Massachusetts—the site of the Battle of Lexington, which precipitated the American Revolution—and enables participants to simulate the activities of a historian. The game functions as a virtual analogue to the Battle and a practice field for historical methodology; it encourages the collection and analysis of evidence, the testing of hypotheses, and formulation of conclusions, in the site where this evidence was first generated. Thus, the participants learn about a specific historic place and time, as well as the context for what occurred there, and construct their own views of the past, while considering alternative views of history (see Figure 3 for a detailed list of pedagogical goals).

The participants’ primary goal in RtR is to reconstruct the events of April 19, 1775 and decide who they think fired the first shot that initiated the Battle of Lexington. To do this, participants walk around present-day Lexington Common and encounter the physical buildings and sites involved in the Battle of Lexington. They also use a personal digital assistant (PDA) to “interact” with virtual historic figures and gather virtual testimonials, evidence, and items, all triggered by Global Positioning Software (GPS) depending on their specific location. For example, when a player approaches the Buckman Tavern (Figure 1), a historical personality such as Paul Revere appears on the PDA and provides his story of the events at Lexington. These virtual historic figures, also called non-playing characters (NPCs), provide a testimonial (and often a document) based on what they think happened before and during the Battle (see Figure 5).

The participants play the game in pairs and as one of four historic roles: Prince Estabrook (African-American slave/Minuteman soldier); John Robbins (free/Minuteman soldier); Ann Hulton (Loyalist/townsperson); Philip Howe (Regular [British] soldier). These participants collect differing evidence based on their historic role in the game; for example, an NPC like Captain John Parker, the leader of the Minutemen, provides very different evidence to a fellow Minuteman soldier than to the female townsperson or the British soldier roles. The game also has two time periods: Time 1 in the game simulates the moment before the

Figure 1. Image of Buckman Tavern in Lexington, Massachusetts



Figure 2. Schematic of RtR; The participants first collect evidence during Time 1 and Time 2, and then compare evidence with other roles during the debate

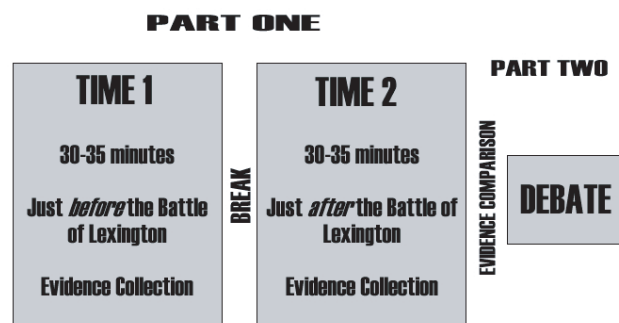


Figure 3. Summary of pedagogical goals for RtR, based on history education standards and John Hoge’s description of historic inquiry (Hoge 2003). The letters next to the goals are referenced throughout the chapter.

(a) Acquire a Meaningful Understanding of Key Historical Themes and People
(1) Understand better the people and leaders involved in the Battle of Lexington and the American Revolution [a1]
(2) Become more aware of the social, economic, geographic, and political context surrounding the Battle of Lexington and the American Revolution [a2]
(3) Learn more about a local historic site and how it functioned in the past. [a3]
(b) Build Knowledge of the Methods and Limitations of History
(1) Question sources and authorial intent of evidence; identify biases in evidence [b1]
(2) Create hypotheses, and draw inferences and conclusions based on historical evidence [b2]
(3) Consider the limits of historical methods and representations of the past [b3]
(c) Confront Multiple Perspectives and Mainstream Interpretations of the Past
(1) Understand and critique master narratives of the Revolutionary War, the Battle of Lexington and history in general [c1]
(2) View, seek out, consider and manage multiple views of the Battle of Lexington and other historic moments [c2]
(3) Reflect on ones’ own perspective on the past and recreations of events [c3]

Battle has begun, and Time 2 simulates the moment immediately after the Battle ends (see Figure 2 for a schematic of the game). The NPCs provide different information in Time 1 and Time 2, and there is a short game break between the two time periods. Then, after the participants gather information in Time 1 and 2, they collaboratively compare their evidence, share hypotheses, and debate who they think fired the first shot.

Augmented Reality (AR) Games for Education?

The development of RtR stems in part from two recent educational initiatives. First, there have been reforms in history standards to include “doing history” activities—such as evidence investigation and validation, exposure to multiple historical views, and narrative creation—and educators are beginning to search for new ways to teach critical thinking as it relates to the study of history (International Society for Technology in Education, 1998;

National Center for History, 1994). Simultaneously, video games are gaining increased acceptance as educational tools and supplements to classroom curricula, and wireless handheld devices such as PDAs are becoming more ubiquitous in the classroom because of their low cost, flexibility, accessibility, wireless capability, portability, and ease of use (Dede, 2004; Dieterle, 2005; Klopfer, Squire, & Jenkins, 2003). Although PDAs are the primary devices discussed in this chapter, cell phones should also be studied as potential platforms for educational games, especially because of their already high penetration in the student population. Thus, I wanted to explore the possibility for using augmented reality games for history education

Augmented reality (AR) games are gaming environments that integrate virtual, location-specific media within the physical world. To play an AR game, participants might use GPS-enabled PDAs or other wireless handheld devices to access virtual information that has been previously mapped to specific locations. For example, a game designer could pre-program information to appear at specific GPS coordinates, or embed information in a specific location that only can be retrieved by reading an RFID (radio frequency ID) tag. In an AR game, participants can use their PDA to interact with real-world objects or receive data about a particular spot in the physical environs. A building or historic site can suddenly become a game board, and statues or doors can provide virtual clues or act as portals in the context of the game.

Thus, AR games may potentially allow students to “do history” situated in a real-world context, rather than passively learn historical “facts” in a classroom. This reflects Brown, Collins, and Duguid’s “situated learning” paradigm, in that the “concept” of history remains married to the activity and culture (1989). AR games may also further encourage collaborative learning because the portability of the devices encourages physical and social interaction, and, for example, the sharing of ideas and collaborative decision-making. Moreover, AR games may motivate those with the increasingly prevalent neo-millennial learning style—learners who favor more experiential, reflective, mentored and collaborative learning, nonlinear expressions of ideas, and individualized learning experiences (Dede, 2005).

AR games, however, do not necessarily support learning, collaboration, imagination, or interest. How do we design educational AR games that are deeply engaging and motivating, while also incorporating history pedagogy? How do we use AR games to teach students to consider both the holistic and microscopic, to negotiate multiple viewpoints, evaluate diverse opinions, and, more broadly, to be more democratically engaged?

In the next sections, I provide an overview of the design process of RtR and then explore in more detail nine elements of the game, and how these elements support the game’s pedagogical objectives and provide deeper immersion in the game world as well as the historic moment of the Battle of Lexington.

Designing RtR

One of the major challenges in designing RtR, and in educational games in general, is incorporating learning goals into the game play: “Play which is not removed from a learn-

ing experience, but inherent to it” (Squire, Jenkins, & The Games-To-Teach Team, 2003, p. 19). How do we embed learning within the game in a way that does not water down the educational content or disengage the player from the fun of playing a game (Thomas, 2003)? Even more challenging, perhaps, is how you do all of this with a limited budget (\$0), limited resources (just myself), and limited time (less than a year).

The simple answer is that I created the game by taking on multiple roles: designer, writer, researcher, tester, and educator. I am not a historian, nor do I have specialized knowledge of the Battle of Lexington, so I immersed myself in the town of Lexington and its lore. The game content—including the testimonials, documents, and game item descriptions were a mix of artistry and historic precision. I needed to balance pedagogical, practical, design, historical, and dramatic concerns to create an engaging experience that was based on accurate portrayals of the viewpoints on the Battle.

I also studied current AR games and considered how to translate them to a historical setting. MIT’s Teacher Education laboratory has developed and tested four AR games, each of which invites participants to solve scientific problems from within an authentic practice field. One, “Environmental Detectives,” is an outdoor AR game where participants work in teams to analyze a virtual oil spill that occurred on the actual MIT campus. The participants navigate a physical location and use a PDA to “interact” with pre-scripted virtual experts and gather information on the toxicity of various predetermined locations around campus (Klopfer, Squire, & Jenkins, 2002). In “Oubreak@MIT,” an indoor AR game, participants work in teams to investigate a simulated disease outbreak on the MIT campus. To investigate the disease, participants use handhelds to interview virtual people in specific rooms around MIT’s campus, obtain and analyze virtual samples, and medicate and quarantine players who might be “infected” with a disease. Similarly, in “River City AR,” a handheld game based on Dede’s MUVE (a multi-user virtual environment where participants can interact with digital artifacts in a 3-D, networked environment), participants investigate a potential biological epidemic in an outdoor portion of MIT’s campus. A team of participants with distinct roles must work together and interact with virtual characters to examine a simulated spread of disease.

MIT’s Teacher Education Laboratory created a prototype of their AR game editor system, which enables designers like me to modify “River City AR” and create a GPS-enabled, location-based, role-playing AR game with little programming. RtR served as a test case for their in-progress editor system, and future versions have since been used to design other educational AR games.

Iterative Design

In creating RtR, I used an iterative design process, which Eric Zimmerman (2003, p. 176) explains as “a design methodology based on a cyclic process of prototyping, testing, analyzing, and refining a work in progress.” With an iterative process, Zimmerman continues, the product or game develops through a meaningful dialogue between the designers and the audience or participants. This means constantly testing and reassessing the game using actual participants, observing their play and asking targeted questions, redesigning the game, and then testing it out again.

I began prototyping elements of the game play a full year before finally testing the full-scale game. For the full-scale tests, I ran two phases of trials. The Pilot study included two trials with a mix of graduate students and educators. Following these results, I did an extensive redesign of the game and then conducted another trial of the game (the Redesign trial), which involved eight individuals, ranging in age from 13 to 17, all attending local and regional high schools. The Pilot and Redesign trials included pre- and post-game surveys on game play, attitudes toward history, and knowledge of the Revolutionary War; videotaped and in-person observations of participants' level of engagement in the game; and a content analysis of the debate, game discussions, and notes. Please see Figure 4 for a comparison of the main game elements in the Pilot and Redesign trials.

Results and Findings

In this section, I present the major results from the Pilot and Redesign trials of the game. In doing so, I describe some of the initial problems with "Reliving the Revolution," and how I redesigned the game to better support engagement and learning, as well as the results of these changes. In what follows, brackets referencing Figure 3 are used to match the game's findings to the original desired learning outcomes. For example, [b2] references the goal: "Create hypotheses, and draw inferences and conclusions based on historical evidence." Hopefully others will find these helpful as suggestions for how to think about creating engaging educational AR games.

Overall Results

The results of my initial and iterative tests suggest that an AR game such as RtR, if designed appropriately, can engage learners in a historic moment and place, and in the practice of history. RtR motivated discovery and enthusiasm; reflection, collaboration, and teamwork; problem solving, interpretation, and analysis; and the consideration of alternative views of history. The participants enjoyed being in the actual site of the Battle of Lexington—the PDAs afforded them the opportunity to integrate virtual information with a real-world context. The game felt novel and authentic, and the participants felt as though they were imbued with a special responsibility to solve an important "history mystery." They took this "serious game" very seriously—they embraced its challenges and critically immersed themselves in the game as such. Accordingly, the participants acted in their historic roles, roles as game players, and roles as learners. By trying on these new identities, striving for a common goal, and collaborating with others, the participants were more open to consider new perspectives, re-evaluate their beliefs and values, and create bridges among them (Gee, 2003). Finally, RtR encouraged the participants to reflect on their interpretations of the event, but also to think more deeply about their preconceived notions about the Battle of Lexington and history in general. The participants began to take their more multifaceted conceptualization of the American Revolution and relate limitations in historical understanding to other situations and even global social issues.

Figure 4. Summary of game elements for the pilot study and redesign study

	PILOT	REDESIGN
Locative technology	GPS	GPS
Game infrastructure	XML/.NET (RiverCityAR Engine)	XML/.NET (RiverCityAR Engine)
Instructions?	In-person instructions by educator	In-person instructions by educator
# of time periods	Two	Two
Linearity?	Nonlinear	Nonlinear with more direction (built-in check points)
Goal	Who fired the first shot at the Battle of Lexington?	Who fired the first shot and two mini-goals for each role
# of roles played	One out of four	One out of four
Pairs?	Play one role as a pair	Play one role as a pair
Collaborative?	During evidence collection, in pairs; during debate as a group	More inter- and intra-pair collaboration throughout
Length of game	~80 min + debate	~80 min + debate
Game play	Search for location-based NPCs and game items triggered by GPS; interview NPCs and gather documents and items	Search for location-based NPCs and game items triggered by GPS; interview NPCs and gather documents and items

Goals and Motivations

Overall, the participants were enthusiastic about the primary goal (who fired the first shot?), and it seemed to motivate their actions throughout the game, structure their navigation of the evidence, and draw them further into the game world. One participant noted that this goal helped orient the way she read and used the evidence, and how she managed a vast amount of game content [a1, a2], saying that “When you have a goal to figure out, you look more.” Her involvement in the game world was more targeted because she had this specific goal in mind.

It was sometimes difficult, however, for participants to balance the extensive game information with the requirements of the game play, including the game’s goal:

I loved the detail, but I was overwhelmed by it. I wanted to take more time to let it sink in... To me it was a push and pull between getting immersed in the detail, and needing to remember to look around me. ... But on the other hand, that is also what makes it rich, because the richness of detail delivers the message that you want—that this is a complex thing, there are lots of points of view, and there was a ton of stuff going on [during the Battle of Lexington].

This comment also points to the difficulty as a game designer in finding a delicate balance between discovery and familiarity. RtR needs to provide enough novelty and a diversity of viewpoints to engage the participants and fit with my educational objectives, but it also needs to imbue participants with a sense of accomplishment. It needs to always be offering new tidbits of data, while also enabling participants to quickly grasp the bigger historic picture of this shared game world, no matter which order the participants actually navigate the game.

Thus, to further direct the participants' navigation of the game and provide more checkpoints in which to measure game progress, I redesigned the game to include smaller objectives—in the form of role-specific secret missions or mini-objectives—which helped the participants break down and compartmentalize the larger historic problem. Explained one participant, the “secret missions kinda orient you to figure out a certain thing.” Echoing this, one participant thought that the question of “who fired the first shot” was too broad, and felt that even more secret missions would make the game “more definite.” Moreover, to complete the secret missions, participants needed to rely on a piece of information only gathered by a different historic role, so participants were more motivated to collaborate with other groups. These mini-objectives, such as naming the spies in the town, or finding out what was in Paul Revere's trunk, also helped the participants piece together the events at Lexington and address the broader question of who fired the first shot. Tackling the primary goal was initially very unwieldy; but after investigating the mini-questions, the participants became more comfortable with their evidence, and were better able to back up their claims, and provide counter arguments to other participants' information [b1, b2].

Game Play Constraints

In any game, it is important for the participant to know how it ends, how success is measured, and what qualifies as “being done.” In “RtR,” the evidence collection period ends when the time runs out, and then the entire game finishes during the debate when the participants agree upon a commonly understood story of the Battle of Lexington and who they think fired the first shot. Thus, it was essential to communicate the time constraints to the participants and ensure that they understood these limitations [b3]. A few participants felt that verbal communication of the time limit was not enough; they also needed visual reminders of the time constraint to motivate them, particularly because they were not competing against each other, but against the clock. Future iterations of this game should have a countdown clock incorporated into the interface of the game or a different visual or audio reminder of time running out.

No participant can possibly gather all the available historic evidence in the time allotted. This constraint helped the participants better appreciate the limits of interpreting the past without all the possible evidence, a key learning objective of the game [b3]. Likewise, during the debate period, many participants acknowledged holes in their data or the need for specific material that would help them feel more confident in their conclusions. For example, one Redesign study participant said the following during the game debate:

How do we know for sure? I feel like we don't have enough evidence. Even with all of us finding different stuff, finding different things, pieces of evidence, how do we know who fired the first shot? Of course if you are loyal to British ... you are going to say, 'Oh the Minute-men fired the first shot' and if you are loyal to America and you are fighting the British, you are going to say, 'the British fired the first shot.' So how do you know?

As the participants began to reflect on their ability to construct valid narratives of the past, they also grappled with current political issues, and how these issues would be later reflected in the history books [c1, c2]. Said one participant:

In America, we have American textbooks and they are written by Americans, so of course you always get that portrayal of the British as being the bad guys and I'm sure the British kids when they learn about this, it's completely different. ... The same with Iraq, people are going to, in years to come when we read about that in textbooks, it is going to be different.

Such a critique on history construction stemmed in part from the *lack* of constraints in this game: participants were liberated from focusing on the single point of view that is often present in textbooks. I carefully devised the game's content to incorporate multiple, alternative views of the past instead of one restrictive master narrative; and I encouraged the participants to create their own interpretations of the past. This freed the participants to try on others' perspectives of past and current moments, and consider how their own cultural and socioeconomic status affects their point of view [c3].

Collaboration and Social Interaction

Almost all of the participants enjoyed playing the game with a partner because they could share ideas and tasks, engage in mini-debates, remember information better, practice decision-making skills, and reflect more deeply on the evidence they gathered. For example, one participant noted that she liked playing with a participant because she “could exchange ideas, notes, plan what to do next,” while another participant noted that, “It was fun to play with others, one, to have someone to help with the handheld/taking notes, and two, just to have someone to bounce ideas/theories off.” Because the game play and collaboration necessitated dialogue and the sharing of evidence, the participants needed to reflect on the evidence and their interpretations, formulate and offer hypotheses, and collectively decide on next steps—all activities related to developing critical thinking, collaborative, and problem solving skills [b1, b2].

During both the evidence collection and debate periods of RtR, the four roles can share information. The relatively compact area of the Lexington Common, coupled with the mobility of the handheld, allowed for physical interactions and the verbal exchanges of evidence among the various roles. Participants in different roles could ask each other for advice, share discoveries, and make connections based on serendipitous exchanges. The mini-objectives in the secret missions also further enhanced inter-role collaboration.

Not only did the participants become more deeply invested in the game because of the interdependence of roles and sharing of responsibilities, but also they were engaged in the game play; enthusiastic about its content; and retained, learned, and even taught each other its historical information.

Role-Playing

In each trial, a pair of participants played one of four historic roles during the game. Based on their role, they received distinct information from the NPCs and accessed unique descriptions of game items. This meant that they received slightly biased accounts of the Battle of Lexington, depending on their historic role. In general, the inclusion of the roles further engaged the participant in a historic moment, and also the game play, because they instilled them with the responsibility of seeing Lexington through the eyes of another.

My challenge was designing the game to ensure that the participants were appropriately immersed in their roles, despite the technical limitations of the game system. First, I was restricted by the River City AR game system because although the participants could receive information from NPCs, they could not actively interact with NPCs *in* their roles. In other words, the participants could not *play* their role except with each other. This made it difficult to emphasize each participant's unique role in the game, as they could not fully test out their new identities. Therefore, I needed to design the game content ahead of time and use language, tone, and style to reinforce to the participants the characteristics of their historic role.

The initial results were mixed. Some of the Pilot study participants commented that during the game they forgot they were playing a historic role, and felt as though they were instead applying a perspective as a filter on the information they were receiving. Other participants, however, felt that the roles were very dynamic, and felt that having a role helped them better understand alternative perspectives on the Battle. For example, one participant stated that it helped him realize "that you cannot just take one point of view when trying to understand and re-creating historical events." Another said it was "interesting to learn the different characters' reactions to the players' roles;" similarly, another participant felt that having a role was "engaging, I felt like I got a lot of information from other characters." Some participants expressed a personal allegiance to the perspective of their historic role, and tended to credit evidence that supported their views. Moreover, during the debate period, when offering evidence to support an argument, each participant considered their role as integral in their analysis and estimation of each piece of evidence [b1, b2].

The participants felt more invested because they had experienced the moment of April 19, 1775 from a distinct point of view, and had gathered evidence accordingly [a1]. They were also more motivated to fulfill the requirements of the game because they had developed loyalty to other game participants and to their role; they relied on each other to interpret evidence or find the next hot spot. Furthermore, sharing a role with a partner provided a point of commonality, compelling them to work more closely together and initiate dialogue for the evidence they gathered. And because pairs needed to compare and corroborate evidence with other pairs, they had to collaborate and collectively seek out other perspectives [c2].

In the redesign of the game, I further incorporated the benefits and limits of each historic role into the game play. For example, the Prince Estabrook role was not able to talk to as many historic figures; however, he was privy to certain testimonials, which emphasized his status as well as his role as a game character. I also created role-specific tasks (the mini-objectives, described previously), in addition to the main objective of figuring out who fired the first shot, to make the roles even more interdependent. Finally, I provided to each participant a physical nametag with the name of the role and type of role to again emphasize that they were playing a role and that this affected their game play.

Integration of Physical and Virtual Worlds

The integration of text and images with the physical world of Lexington seemed to help the participants absorb, categorize, remember, and recall the information, and supported multimodal learning styles. The overlap between the physical and virtual seemed to motivate and direct the navigation of the historical content, while encouraging deeper inspection of the physical environment [a1, a3]. When the virtual information supported what was being seen in the environment, it seemed more valid. For example, when participants read John Parker’s testimonial and saw that it was echoed on a small monument on the Lexington Common, they regarded it as more reliable [b1, b2].

The interplay among the real, the virtual, and one’s own imagination also created location-based mini-stories, which further engaged the participants in the overall “story” of Lexington. Said one participant:

My favorite part of the game was when I found out that John Harrington had been shot in the game. It was neat because you met someone on the street who told you that [he was dead] and then you ran into his wife who was like ‘Oh my god.’ And then you passed his house and you looked at the house and it has the actual plaque saying that this is where John Harrington died in his wife’s arms and it corresponded to the story. And then you saw John Harrington, who was dead and didn’t have anything to say, and it felt like the process of discovering.

Thus, participants could access and interact with these spontaneous mini-stories, which were amalgams of physical and virtual narrative threads. This further strengthened the connection between the real and virtual worlds, and also helped the participants to construct rich narratives about their game experience, leading to greater retention of information and enjoyment of the game.

For example, one participant in the Redesign trial explains why he felt that this game helped him learn about the Revolutionary War better than a traditional history classroom:

I re-learned U.S. History One, which is what I took sophomore year of high school, and it was a total waste of my time. And I just re-learned it in three hours...this recapped it and I re-learned it and now I know more about history... the pictures, and the items [helped make it clearer].

The richness and variety of the game content, coupled with the ability to explore a physical site, contributed to the participants' interpretive dialogue throughout the game, and especially during the debate period, where they provided detailed arguments and in-depth syntheses of information [b1, b2]. The location-anchored historic data quickly became tangible building blocks with which to construct and share a narrative of Lexington with others.

By combining a present-day physical environment with a virtual historic moment, RtR enabled a deeper exploration of the historic site of Lexington, Massachusetts, while effectively conveying the Battle of Lexington from diverse and reflective perspectives. The interplay between the physical and virtual deepened the participants' engagement with both worlds, while also creating unique connections.

Reflection and Debate

In RtR, after gathering evidence for an hour, the participants all come together to collectively debate what they think happened in Lexington in 1775. This debate period is an important extension of the learning because a major pedagogical goal of the game is that participants are not just gathering material, but using it: sifting through it, analyzing it, applying it, talking about it, and showing where they retrieved it. Also important was providing time so that the participants could reflect on their process to solidify new knowledge and further integrate historic methodology and concepts. The debate period was also a motivating factor for the participants in the game, because it further compelled them to learn their roles, thoroughly and intelligently gather evidence, and interact with the game world. It was an important impetus for the participants, because they knew they would have to share their findings and lessons learned with the group later, and that the other participants would rely on them for knowledge.

In each of the trials, the participants created unique hypotheses and interpretations of the material; constructed counter arguments; and worked fluidly among multimodal texts. Participants were constantly referencing and questioning evidence—sharing it verbally and physically showing it to each other on their handhelds. They would read aloud pieces of testimonials and offer hypotheses or counter-arguments based on these stories or documents [b2, c2]. When providing evidence, they considered the source of the evidence, as well as their own historic role and his/her relationships with the NPCs [b2, c3]. Each participant seemed fully invested in trying to determine who fired the first shot at Lexington's battle based on their evidence, and each trial of the game had a distinct final conclusion based on the consensus of the group [c1].

Although the participants were able to grapple with a large amount of information, they sometimes needed targeted questions and suggestions to guide their debate, encourage reflection, or properly contextualize their evidence. The participants needed scaffolding to support their visit to this "practice field" and the incorporation of new epistemic frames, tools, and concepts. This further expresses the need for an instructor or mentor to direct learning in the game and that multimedia platforms are not replacements for teachers.

The debate and collaborative decision-making during the game seemed beneficial to strengthening the participants' understanding of history, their application of critical thinking skills, and reflection on historical inquiry, but it also made "RtR" sometimes feel less like a "game."

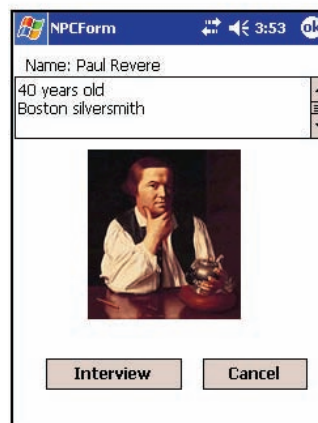
One participant in the Pilot study noted that while she loved the collaborative aspect, she felt like, since there was no competition *per se*, “we’re all going to just come together and see what we got anyway.” Thus, it did not feel as pressing to gather everything, or to digest all the evidence, since there would be time for sharing later. In the Redesign study, I further emphasized that the participants, and their concomitant roles, were dependent on each other to gather more information or solve the mini-objectives, and this seemed to increase the participants’ motivation to collect and interpret evidence.

Overall, the debate period enabled the participants to retrace their steps; review and apply the game’s historic content; and practice teamwork, hypothesis formation, and analytical skills. Moreover, it gave the participants an opportunity to reflect on their own processes of evidence appraisal and history construction, and relate their game experiences to others.

Nonlinearity and Control

The nonlinear structure of the game, which allows the participants to interact with historic figures and access their stories in any order, evokes temporal simultaneity—which I hoped would underscore the idea of multiple truths and possibilities. In other words, the self-guided navigation of the game and open availability of the stories further suggests that one view is not necessarily more correct than another. In the trials, the participants considered alternative perspectives of the Battle of Lexington, for example, said one participant, “I learned about all the different sides. Normally you would just think of the American soldiers and the British soldiers, slaves, the wives, ... the Minutemen, there are people frustrated here for personal reasons, patriotic reasons ... You get a sense of the different roles of that time period.” Furthermore, the game’s nonlinearity also more closely mimics the work of a historian. No historian has a set linear path in which to gather evidence; s/he must navigate a vast archive of historical information and create his/her own version of the past.

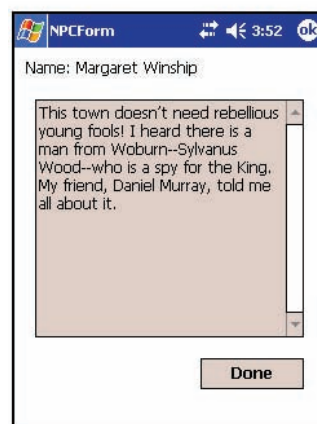
Figure 5. Screenshot of a virtual historic figure (NPC); participants can click on “interview” to receive a testimonial from Paul Revere



Some participants, however, desired a more linear game. They felt that because the game was open-ended, it did not feel as goal-oriented. They were unsure of their status in the game as they navigated it because there was little feedback on whether they were gathering enough evidence or finding enough historic figures. Some felt that there were too many choices at once and they wanted more of a progression of events.

In the Redesign, I established new mini-objectives as role-specific secret missions (as described previously). These objectives functioned as checkpoints that helped direct the participants' navigation of the content. Interestingly, the Redesign trial participants especially did not seem overwhelmed or frustrated, but thrived in the nonlinear environment because they liked to "figure something out for [themselves]." These participants enjoyed having agency or control over how they navigated the game world, perhaps because these younger participants appreciated the opportunity to transgress boundaries during learning, since traditional education is usually highly structured (Gee, 2003). For example, one participant liked that in this game, the results were not pre-established and she needed to create the "game ending" herself. Through a self-directed construction of her learning, she was able to delve more deeply into the historic moment, as well as the game itself. She said, "in [this game] you had to put it together, you had to research and then figure something out for yourself. It wasn't like a set [answer] like 'you have to click on this conclusion now.' You have to come up with whatever." Having the responsibility to perform motivated her to complete the game and engage in its world (Oblinger, 2004), but then to also question what she received, and devise her own, novel interpretation of the past [b2, c1, c2]. A nonlinear world with well-placed boundaries and sufficient direction seemed to support the pedagogical objectives of encouraging alternative views of history, and also engaged participants by enabling them to have enough control over their game experience to encourage them to use their critical thinking skills as well as their imagination.

Figure 6. Screenshot of a testimonial from Margaret Winship, a virtual historic figure (NPC), found in the time period before the Battle of Lexington



Mobility and Discovery

Most participants felt that the ability to wander around the site of Lexington and discover information at particular sites was one of their favorite aspects of the game. One participant noted that walking around made the game “more engaging and more interactive,” and another said that the best part of the game was “getting to know a physical site. To me that is the excitement of the game.” [a3]

Being outside in a group playing a game and exploring a site together also facilitated more social interactions, which the participants felt added to their enjoyment of the experience. The physical nature of the handheld game may also have increased the game’s collaborative potential—being close together and crossing each other’s paths encouraged the participants to interact socially.

This game was especially appealing to kinesthetic learning styles. Many participants found particular pleasure in moving around the Lexington site and discovering information embedded in the environment, especially as opposed to sitting in the classroom and being taught the same information, as exemplified by the following exchange:

Participant 1: *Yeah, if we sat in a classroom and did this and I would walk away and be like “Yeah, okay.”*

Participant 2: *But when you are actually moving around to do it ... I think it’s definitely more interesting to do it this way than to sit in the classroom.*

In addition, the novelty of watching the PDA suddenly present an NPC or game item “mimicked the process of discovery,” as one participant commented. Another participant noted that, “the thing we had the most fun with on our team was ‘oh we’re getting closer, oh we found one,’ the wandering and finding” of the hot spots, which contributed to greater engagement in the game, and thus, increased curiosity about its content.

Authenticity

Integral to simulating and teaching history is the creation of an authentic learning environment for practicing historical methodology. In RtR, the participants were in the place where the Battle of Lexington occurred; they were viewing the real historic buildings and structures mentioned in many testimonials; and they were reading detailed personal accounts of the Battle from actual historic figures—all of which provided a distinct and immersive experience of the historic moment. This feeling of authenticity underscored all of the game’s activities, and strengthened the connection between game play and historical methodology. Said one participant, “[The game] put you in the real place where everything happened. It gave you the real, actual people who were there, like the names and their opinions.” [a1, a2] In RtR, participants are evaluating “real” evidence in a “real” place, and attacking a “real” question based on a “real” battle, which encouraged them to practice and apply critical thinking

skills. Almost every participant enjoyed being at the site of Lexington because they felt that it “made history more real” and gave them a better sense of the “history of the site.” [a3]

The ability to apply historical inquiry skills in an actual historical site also added to the authenticity of the game’s tasks, their desire to complete them and learn from them, and their willingness to apply this learning to other similar tasks in the future. The participants consistently treated the historical evidence they gathered as valid, and made informed hypotheses, partially because they viewed the historical problem of trying to understand who fired the first shot as authentic. Moreover, the genuineness of their experience further engaged them in the historic moment of the Battle of Lexington, and also in historical methodology and construction. They felt responsible for deciding who fired the first shot, so they were more passionate about playing the game. They felt like historians; therefore, they took the game’s tasks more seriously and performed the related critical thinking skills more rigorously.

Design Summary

The following is a list of recommendations for educational AR games (and games in general), as derived from my process of simultaneously designing RtR to meet my pedagogical objectives and engage the participant.

- **Goals:** Participants need feedback and “checkpoints” throughout the game, so include both large overarching goals as well as smaller mini-objectives. The large ones guide the game play to the ending and help set boundaries, while the smaller ones can heighten their sense of engagement, responsibility, and accomplishment, and help make the primary one more manageable.
- **Constraints:** Provide necessary constraints (and reminders of constraints) to motivate game play, particularly in non-competitive games. Also, consider well-timed or well-placed open-endedness in the game play to encourage participants to experiment with new ideas or identities.
- **Collaboration:** Enable social interaction by placing participants in groups, teams, or roles and creating tasks that require discussion, debate, or the sharing of resources. Also, provide distinct game content or game play to various teams or pairs, encouraging them to exchange information or allocate responsibilities.
- **Roles:** Include roles to further engage the participants in the game’s content, and also increase their investment in the game. Provide distinct information, responsibilities, and goals to each role to create more interdependence among participants and heighten motivation.
- **Integration of Physical and Virtual Worlds:** Consider how the real world and virtual information can interact to spark unique connections, enhance learning, and further engage the participants in both the game site and its educational content. Think about how the interplay between the real, the virtual, and one’s imagination can create powerful stories and help participants more fully integrate new knowledge.

- **Reflection and Application:** Provide time in the game not only to absorb information, but to apply it in ways that enhance the game’s pedagogical goals. Allow time for reflection; while there should be periods of novelty and discovery, there should also be periods that simply strengthen connections to the new material.
- **Control and Navigation:** Find ways to enable the participant to personalize his/her experience and feel a sense of agency over one’s game destiny, while also creating supportive boundaries and checkpoints so that participants are not too overwhelmed. Respect the participant as an equal—encourage them to make their own decisions, test rules, and guide their own game play. Enable unexpected permutations and juxtapositions, as you will never be able to design for every possible scenario.
- **Mobility:** Particularly in AR and location-based games, find ways to motivate a deeper discovery of a physical site, while also encouraging a “directed wandering.” Consider how mobility and physical interactions can support greater collaboration, and more serendipitous social exchanges.
- **Authenticity:** Think about how to create a game environment that appropriately mimics or simulates the processes you want to teach. Provide real data, evidence, sources, names, places, and people when you can, and think of the game as a “practice field” where students can actually perform skills or tasks.

Next Steps

As I continue to refine RtR, larger questions of choice emerge—how we balance practical, artistic, pedagogical, and historical considerations to create such a game—and what that means for how we represent a historic moment. Taking it to the next level: how do we encourage the metacognition of these educational games, and enable kids to assess a game’s implications for conveying history and historical thinking? My response would be to encourage students to create their own mobile games, and to reflect on their own game design decisions. Using these general principles as guides, they could choose a local and/or personally meaningful site; work together to research historic figures and write testimonials; outline and experiment with the game’s play; and then analyze how these choices affect and reflect our understanding of history. By becoming creators of the medium, they will be even better able to evaluate critically their own and others’ assumptions about history. These learners can then take their newly prismatic eye and apply it to other disciplines—to identify biases in newspapers, consider authorial intent in an essay, privilege information on one Web page versus another, and incorporate other perspectives in a debate.

Activities such as RtR are essential for further study because they can potentially engage learners not only as participants in a game, but also as more active participants in society. As Jefferson suggested during America’s infancy, history is integral to citizenship (Carpenter, 2004)—and this has become more important as America becomes more economically, socially, and culturally varied. By understanding the past, people can better evaluate the future; by listening to multiple perspectives, people are more empowered to appreciate a situation, and better equipped to defend their freedoms (Carpenter, 2004).

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